

Stein's Lemma, Optimism, and Model Complexity

A complete end-to-end derivation

Lecture notes companion — Autoencoders & Regularisation

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Abstract

These notes give a fully self-contained derivation of the *optimism* of training error and its connection to model complexity through **Stein's Lemma**. We start from the data-generating process, prove Stein's Lemma from first principles, apply it to the regression “noise–prediction” covariance, interpret the resulting derivative as *degrees of freedom*, and finally work out the clean closed form for ordinary linear regression, where the complexity penalty equals exactly the number of parameters p . We close with the “master equation” that underlies all of regularisation.

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1 Setup: the data-generating process

We assume the observed labels are a true signal plus i.i.d. Gaussian noise:

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n. \quad (1)$$

- $f(x_i)$ — the *true* (unknown) signal, a fixed number.
- ϵ_i — the *noise*, with $\mathbb{E}[\epsilon_i] = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$.

- y_i — the *observed* data point.

The model is fit on $(y_1, \dots, y_n)$, producing fitted values

$$\hat{y}_i = \hat{f}(x_i). \quad (2)$$

Remark 1 (Why does the noise affect the fit?). *Because the model is trained on the y_i 's, and each y_i carries its noise ϵ_i , the fitted function is indirectly a function of all the noises:*

$$\hat{f} = \hat{f}(\epsilon_1, \dots, \epsilon_n).$$

This dependence is exactly what makes ϵ_i and $\hat{f}(x_i)$ correlated, and is the whole reason the optimism term is nonzero.

2 Train error, test error, and optimism

The two empirical errors are

$$\text{trainerr} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2, \quad \text{testerr} = \frac{1}{m} \sum_{i=n+1}^{n+m} (y_i - \hat{f}(x_i))^2. \quad (3)$$

What we actually care about is the *true error* $\mathbb{E}[(\hat{f}(x_i) - f(x_i))^2]$, which we cannot compute directly because f is unknown. Expanding $\hat{y}_i = \hat{f}(x_i)$ and $y_i = f(x_i) + \epsilon_i$,

$$\underbrace{\mathbb{E}[(\hat{f}(x_i) - f(x_i))^2]}_{\text{true error}} = \underbrace{\mathbb{E}[(\hat{y}_i - y_i)^2]}_{\text{empirical error}} - \mathbb{E}[\epsilon_i^2] + 2 \mathbb{E}[\epsilon_i(\hat{f}(x_i) - f(x_i))]. \quad (4)$$

The last term is a covariance between ϵ_i and the residual signal (recall $\mathbb{E}[\epsilon_i] = 0$, so $\text{Cov}(\epsilon_i, Z) = \mathbb{E}[\epsilon_i Z]$).

Test data (Case 1). Test points did not participate in training, so $\epsilon_i \perp (\hat{f}(x_i) - f(x_i))$ and the cross term *vanishes*:

$$\mathbb{E}[\epsilon_i(\hat{f}(x_i) - f(x_i))] = \mathbb{E}[\epsilon_i] \mathbb{E}[\hat{f}(x_i) - f(x_i)] = 0.$$

Hence test error tracks true error up to a small constant — the reason we use a validation set.

Training data (Case 2 — optimism). On training data, ϵ_i *was* used to fit $\hat{f}$, so the cross term is positive and the training error is *optimistic* (smaller than the true error). Our goal is to quantify this term:

$$\boxed{\frac{1}{n} \sum_{i=1}^n \mathbb{E}[\epsilon_i(\hat{f}(x_i) - f(x_i))]} \quad (5)$$

Since $f(x_i)$ is fixed and $\mathbb{E}[\epsilon_i f(x_i)] = f(x_i) \mathbb{E}[\epsilon_i] = 0$,

$$\mathbb{E}[\epsilon_i(\hat{f}(x_i) - f(x_i))] = \mathbb{E}[\epsilon_i \hat{f}(x_i)]. \quad (6)$$

Everything therefore reduces to evaluating $\mathbb{E}[\epsilon_i \hat{f}(x_i)]$, and this is exactly where Stein's Lemma enters.

3 Stein's Lemma

Lemma 1 (Stein, 1981). *Let $Z \sim \mathcal{N}(0, \sigma^2)$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with $\mathbb{E}[g'(Z)] < \infty$. Then*

$$\boxed{\mathbb{E}[Z g(Z)] = \sigma^2 \mathbb{E}[g'(Z)]}. \quad (7)$$

3.1 Proof of Stein's Lemma

Let ϕ denote the centred Gaussian density,

$$\phi(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/(2\sigma^2)}. \quad (8)$$

Step 1 — write out the expectation.

$$\mathbb{E}[Z g(Z)] = \int_{-\infty}^{\infty} z g(z) \phi(z) \, dz. \quad (9)$$

Step 2 — the key Gaussian identity. Differentiating the density,

$$\frac{d}{dz} e^{-z^2/(2\sigma^2)} = -\frac{z}{\sigma^2} e^{-z^2/(2\sigma^2)} \implies \phi'(z) = -\frac{z}{\sigma^2} \phi(z), \quad (10)$$

which rearranges into the crucial relation

$$z \phi(z) = -\sigma^2 \phi'(z). \quad (11)$$

This is the only place σ^2 is introduced; it survives all the way to the final result. Substituting (11),

$$\mathbb{E}[Z g(Z)] = -\sigma^2 \int_{-\infty}^{\infty} g(z) \phi'(z) \, dz. \quad (12)$$

Step 3 — integration by parts. With $u = g(z)$, $dv = \phi'(z) \, dz$ (so $du = g'(z) \, dz$, $v = \phi(z)$):

$$\mathbb{E}[Z g(Z)] = -\sigma^2 \left[g(z) \phi(z) \right]_{-\infty}^{\infty} + \sigma^2 \int_{-\infty}^{\infty} g'(z) \phi(z) \, dz. \quad (13)$$

Step 4 — boundary term vanishes. The Gaussian density decays faster than any polynomial grows, so $g(z)\phi(z) \rightarrow 0$ as $z \rightarrow \pm\infty$, giving $\left[g(z)\phi(z) \right]_{-\infty}^{\infty} = 0$. Therefore

$$\mathbb{E}[Z g(Z)] = \sigma^2 \int_{-\infty}^{\infty} g'(z) \phi(z) \, dz = \sigma^2 \mathbb{E}[g'(Z)]. \quad \blacksquare \quad (14)$$

What is σ^2 ?

The σ^2 in Stein's Lemma is precisely the **variance of the Gaussian noise** $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. It enters through the identity $z\phi(z) = -\sigma^2\phi'(z)$ and never leaves. Intuitively: if $\sigma^2 \approx 0$ there is little noise to overfit; if σ^2 is large, fitting noise is more costly — which is why the optimism scales with σ^2 .

4 Applying Stein's Lemma to regression

Fix every error except ϵ_i and set

$$Z = \epsilon_i, \quad g(\epsilon_i) = \hat{f}(x_i). \quad (15)$$

Stein's Lemma (Lemma 1) gives immediately

$$\mathbb{E}[\epsilon_i \hat{f}(x_i)] = \sigma^2 \mathbb{E} \left[\frac{\partial \hat{f}(x_i)}{\partial \epsilon_i} \right]. \quad (16)$$

4.1 From a noise derivative to a data derivative

Because $y_i = f(x_i) + \epsilon_i$ and $f(x_i)$ is constant,

$$\frac{\partial y_i}{\partial \epsilon_i} = \frac{\partial (f(x_i) + \epsilon_i)}{\partial \epsilon_i} = 1. \quad (17)$$

(Compare $b = a + c \Rightarrow db/dc = 1$.) By the chain rule,

$$\frac{\partial \hat{f}(x_i)}{\partial \epsilon_i} = \frac{\partial \hat{f}(x_i)}{\partial y_i} \frac{\partial y_i}{\partial \epsilon_i} = \frac{\partial \hat{f}(x_i)}{\partial y_i}. \quad (18)$$

Substituting into (16),

$$\mathbb{E}[\epsilon_i \hat{f}(x_i)] = \sigma^2 \mathbb{E} \left[\frac{\partial \hat{f}(x_i)}{\partial y_i} \right]. \quad (19)$$

4.2 The optimism formula

Summing over i and dividing by n yields the boxed identity from the lecture notes:

Optimism via Stein's Lemma

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[\epsilon_i (\hat{f}(x_i) - f(x_i))] = \frac{\sigma^2}{n} \sum_{i=1}^n \mathbb{E} \left[\frac{\partial \hat{f}(x_i)}{\partial y_i} \right]. \quad (20)$$

5 Interpretation: degrees of freedom

The derivative $\frac{\partial \hat{f}(x_i)}{\partial y_i}$ answers:

“If I perturb one training label y_i slightly, how much does the fitted prediction at point i change?”

This is a direct measure of **model sensitivity**.

Model	Predictor	$\partial \hat{y}_i / \partial y_i$
Simple (mean)	$\hat{y}_i = \bar{y} = \frac{1}{n} \sum_j y_j$	$\frac{1}{n}$ (small)
Complex (memorise)	$\hat{y}_i = y_i$	1 (large)

A simple model barely reacts to a single label; a complex model can change a lot. Summing these sensitivities defines the **degrees of freedom**:

$$\text{df} = \sum_{i=1}^n \frac{\partial \hat{y}_i}{\partial y_i}. \quad (21)$$

With this, the optimism (20) becomes

$$\text{Optimism} = \frac{2\sigma^2}{n} \text{df}, \quad (22)$$

(the factor 2 coming from the $2 \times$ cross term in (4)), explaining why more complex models enjoy lower training error but suffer a larger train–test gap.

6 Closed form: ordinary linear regression

This is the clean example where everything is computable in closed form.

6.1 The hat matrix

For linear regression $\hat{y} = X\hat{\beta}$ with the least-squares solution

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \quad (23)$$

we get

$$\hat{y} = X(X^\top X)^{-1} X^\top y = Hy, \quad \boxed{H = X(X^\top X)^{-1} X^\top}. \quad (24)$$

H is called the **hat matrix** because it “puts the hat on y ”: $y \mapsto \hat{y}$.

6.2 Why the diagonal entries appear

Writing $\hat{y}_i = \sum_j H_{ij} y_j$ and differentiating with respect to y_i , every term vanishes except the $j = i$ term:

$$\frac{\partial \hat{y}_i}{\partial y_i} = \frac{\partial}{\partial y_i} \left(\sum_j H_{ij} y_j \right) = H_{ii}. \quad (25)$$

Note that $H_{ij} = \partial \hat{y}_i / \partial y_j$ answers a *different* question (how point j influences prediction i); Stein’s Lemma needs the *same-index* derivative, hence the diagonal H_{ii} .

6.3 Trace equals number of parameters

Summing the diagonal sensitivities gives the trace,

$$\text{df} = \sum_{i=1}^n \frac{\partial \hat{y}_i}{\partial y_i} = \sum_{i=1}^n H_{ii} = \text{tr}(H). \quad (26)$$

Using the cyclic property of the trace, $\text{tr}(ABC) = \text{tr}(BCA)$,

$$\text{tr}(H) = \text{tr}(X(X^\top X)^{-1} X^\top) = \text{tr}((X^\top X)^{-1} X^\top X) = \text{tr}(I_p) = p. \quad (27)$$

Closed-form complexity for linear regression

$$\text{df} = \text{tr}(H) = p \quad (\text{the number of parameters}), \quad \text{Optimism} = \frac{2\sigma^2 p}{n}. \quad (28)$$

This is the clean conceptual bridge:

$$\boxed{\text{Noise fitting} \rightarrow \text{Sensitivity} \rightarrow \text{Degrees of freedom} \rightarrow \text{Model complexity}.}$$

7 The master equation of regularisation

Combining the pieces from (4) and (20),

$$\underbrace{\text{true error}}_{\text{what we want}} = \underbrace{\text{empirical train error} + \text{small constant}}_{\text{computable}} + \underbrace{\Omega(\text{model complexity})}_{\approx \frac{\sigma^2}{n} \sum_i \partial \hat{f}(x_i) / \partial y_i}. \quad (29)$$

Therefore we should *not* minimise training loss alone but

$$\boxed{\min_{\theta} \left[\underbrace{L_{\text{train}}(\theta) + \Omega(\theta)}_{L(\theta)} \right]}. \quad (30)$$

$\Omega(\theta)$ controls complexity so that the model generalises. This is the foundation of every regularisation method — L_2 weight decay, dataset augmentation, parameter sharing/tying, input/output noise (label smoothing), early stopping, ensembles, and dropout.

The sweet spot

As complexity grows, train error falls monotonically toward 0, but test error follows a U -shape: it first falls (less bias) then rises (more variance). The minimum of the test-error curve — tuned via $\Omega(\theta)$ — is the ideal model complexity. By the Universal Approximation Theorem deep networks are extremely flexible and overfit easily, so regularisation is essential.

8 Appendix: bias–variance decomposition

For a new point $y = f(x) + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$,

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible}}. \quad (31)$$

Proof sketch. Write $y - \hat{f} = (f - \mathbb{E}[\hat{f}]) + (\mathbb{E}[\hat{f}] - \hat{f}) + \epsilon$, expand the square, and take expectations. The cross terms vanish because $\mathbb{E}[\epsilon] = 0$ with $\epsilon \perp \hat{f}$, and $\mathbb{E}[\mathbb{E}[\hat{f}] - \hat{f}] = 0$. Only the three squared terms survive. ■

A simple model has high bias / low variance; a complex model has low bias / high variance — the same trade-off Stein’s Lemma quantifies through degrees of freedom.